Optimizing Computer System Performance

Performance Optimizations

Learning Objectives

Be able to…

- decide where to optimize a software system
- Apply Amdahl's Law and the bottleneck law to predict performance improvements
- Identify when to use common latency and throughput techniques

Modelling a software system

- Think of a system as a collection of stages, each representing a specific operation
  - Each stages has a latency (how fast does input become output)
  - Each stage has a throughput (how many items can it process in a given time)
  - For sequential stages, throughput = 1 / latency
  - For parallel stages, these could be unrelated
- What is the throughput of the end-to-end system given the bandwidth of each stage?
  - The minimum throughput
  - This is called "the bottleneck law"
  - Implication => optimizing for throughput requires optimizing the minimum throughput.
- What is the latency of the end-to-end system given the latency of each stage?
  - The sum of all latencies.
  - Let's imagine that a stage takes p percentage of the overall time and that you c speed it up by a factor of s.
  - Amdahl's Law: the overall speedup is given as $1 / (1 - p + (p/s))$.
    - Intuitively: $1 - p$ is the amount of time for the other stages,
    - p/s is the new time for the stage
  - Implication => focus your efforts on the stages that take a long time!